ET4382 Power Conversion Techniques in CMOS Technology – SMPC Project

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First Calculations

Duty Cycle: $\frac{V_{in}}{V_o} = Duty \;\Rightarrow \frac{1.2}{3.3} = D = 0.3636$

Now, for $I_L = 10mA, with\, V_o = 1.2V, R_L = \frac{V_o}{I_L} = 120\Omega$

Similarly, for $I_L = 10\mu A\ case, R_L = 1.2M\Omega$ (But this will become relevant later)

$F_{sw}, L\ and\ C\ Sizing$

$K = 1 - D = \frac{2LF_{sw}}{R_L}$ (Derivation in Appendix)

So, $LF_{sw} = 38.184$

And, Ripple Current is given by: $\Delta I_L = \frac{(V_{in} - V_o)}{L * F_{sw}} * D$

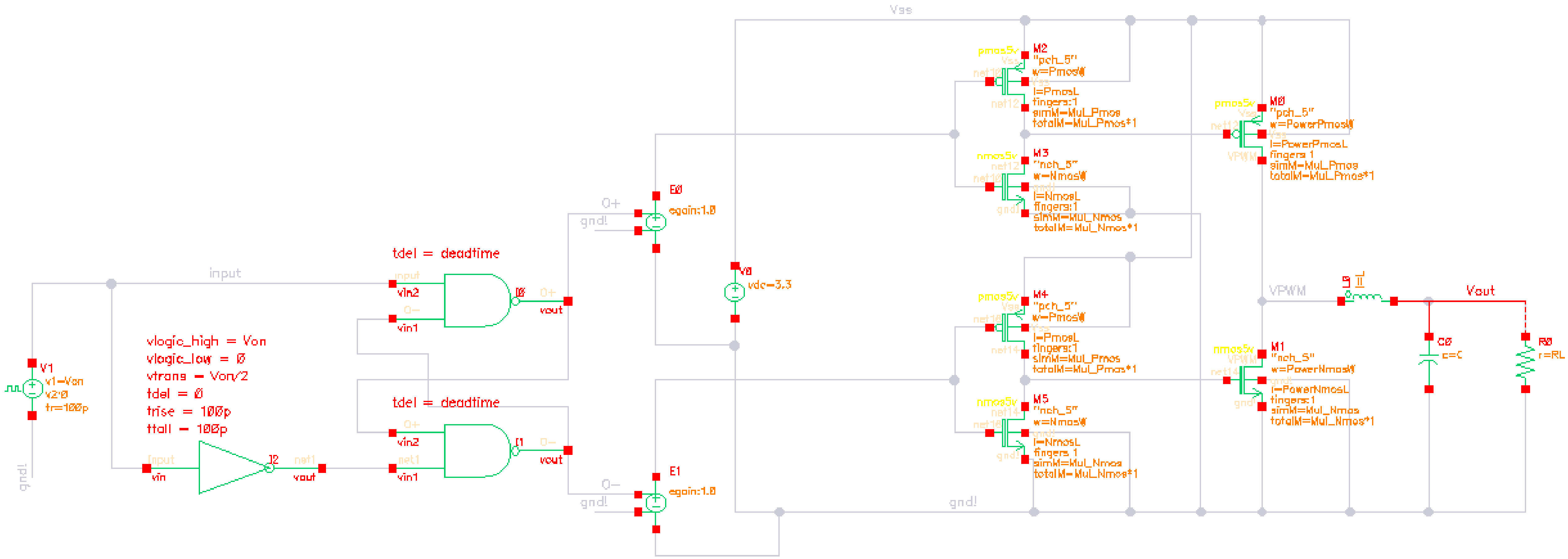
So, $\Delta I_L = 20mA$ (for any combination of $LF_{sw} = 38.184$)

Now, we look at the ripple voltage spec given (100mV):

$I = C\frac{\Delta V}{\Delta T} \Rightarrow C * F_{sw} = 0.2$

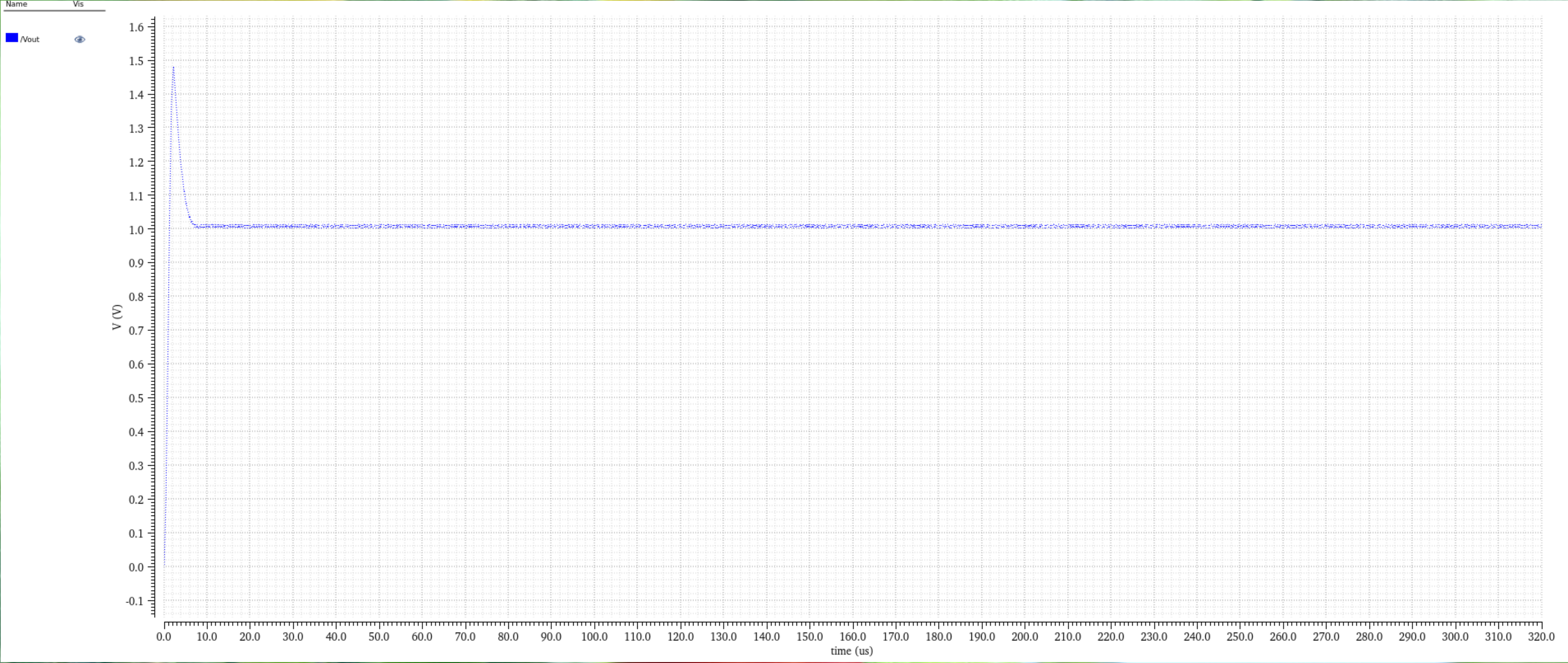
So we have $C * F_{sw} = 0.2$ & $LF_{sw} = 38.184$

Lets set $F_{sw} = 4MHz, so\ C = 50nF\ and\ L = 9.546\mu H$

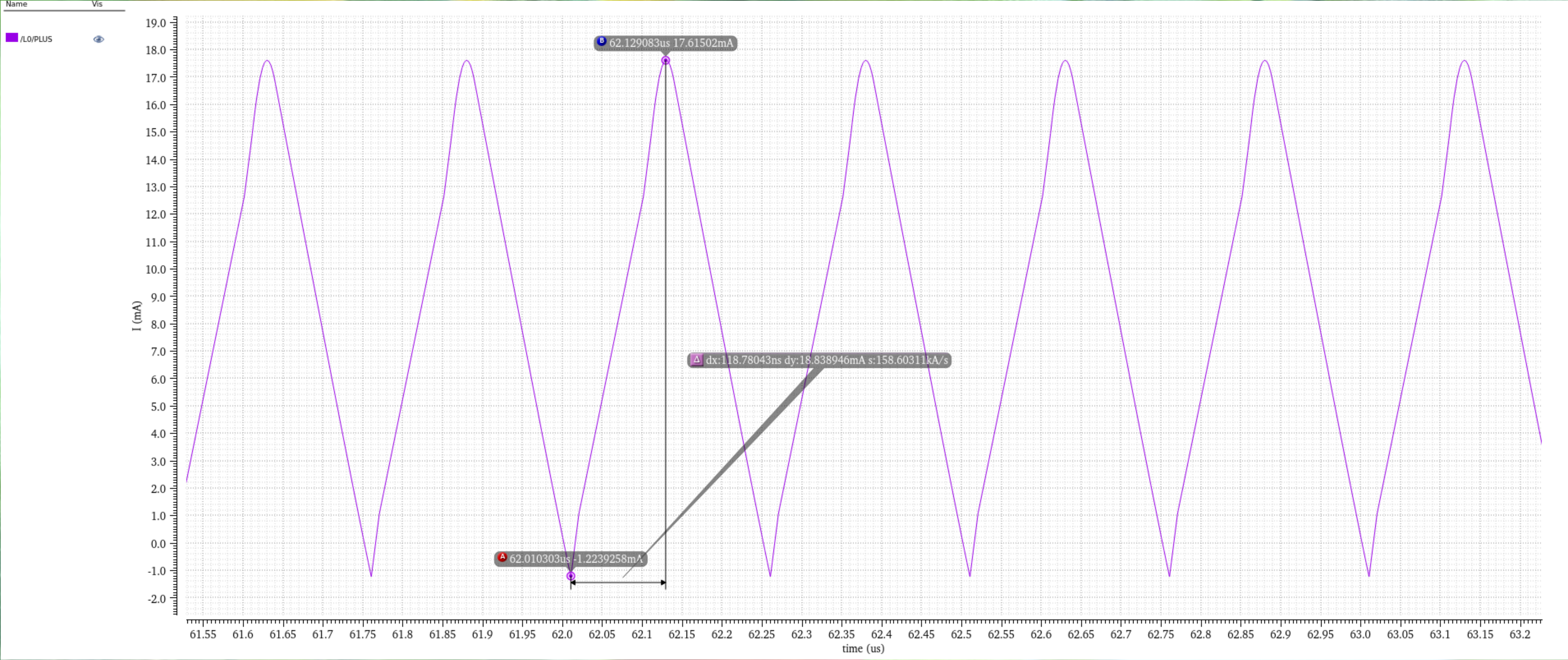


Vss
M2
"pch_5"
w=PmosW
l=PmosL
fingers:1
simM=Mul_Pmos
totalM=Mul_Pmos*1
M0
"pch_5"
w=PowerPmosW
l=PowerPmosL
fingers 1
simM=Mul_Pmos
totalM=Mul_Pmos*1
M3
"nch_5"
w=NmosW
l=NmosL
fingers:1
simM=Mul_Nmos
totalM=Mul_Nmos*1
E0
egain:1.0
gnd!
tdel = deadtime
input
V0
vdc=3.3
M4
"pch_5"
w=PmosW
l=PmosL
fingers:1
simM=Mul_Pmos
totalM=Mul_Pmos*1
VPWM
Vout
vin2
vin1
vout
I0
vlogic_high = Von
vlogic_low = 0
vtrans = Von/2
tdel = 0
trise = 100p
tfall = 100p
V1
v1=Von
v2:0
tr=100p
C0
c=C
R0
r=RL
M1
"nch_5"
w=PowerNmosW
l=PowerNmosL
fingers:1
simM=Mul_Nmos
totalM=Mul_Nmos*1
tdel = deadtime
M5
"nch_5"
w=NmosW
l=NmosL
fingers 1
simM=Mul_Nmos
totalM=Mul_Nmos*1
I1
I2
vin
vout
E1
egain:1.0
gnd!

Schematic



Vout Plot



Ripple Current Plot

Power FET and Gate Driver Sizing

Size the power FETs to trade off switching loss and conduction loss for high peak efficiency

For that, we do a calculation for $P_{cond} = P_{avg} + P_{ripple}$

For finalized W/L values for Nmos and Pmos: 400u/600n and 400u/500n

Calculated Ron values for Nmos and Pmos are 203m and 72m respectively

So, Calculated Efficiency is = 99.3%

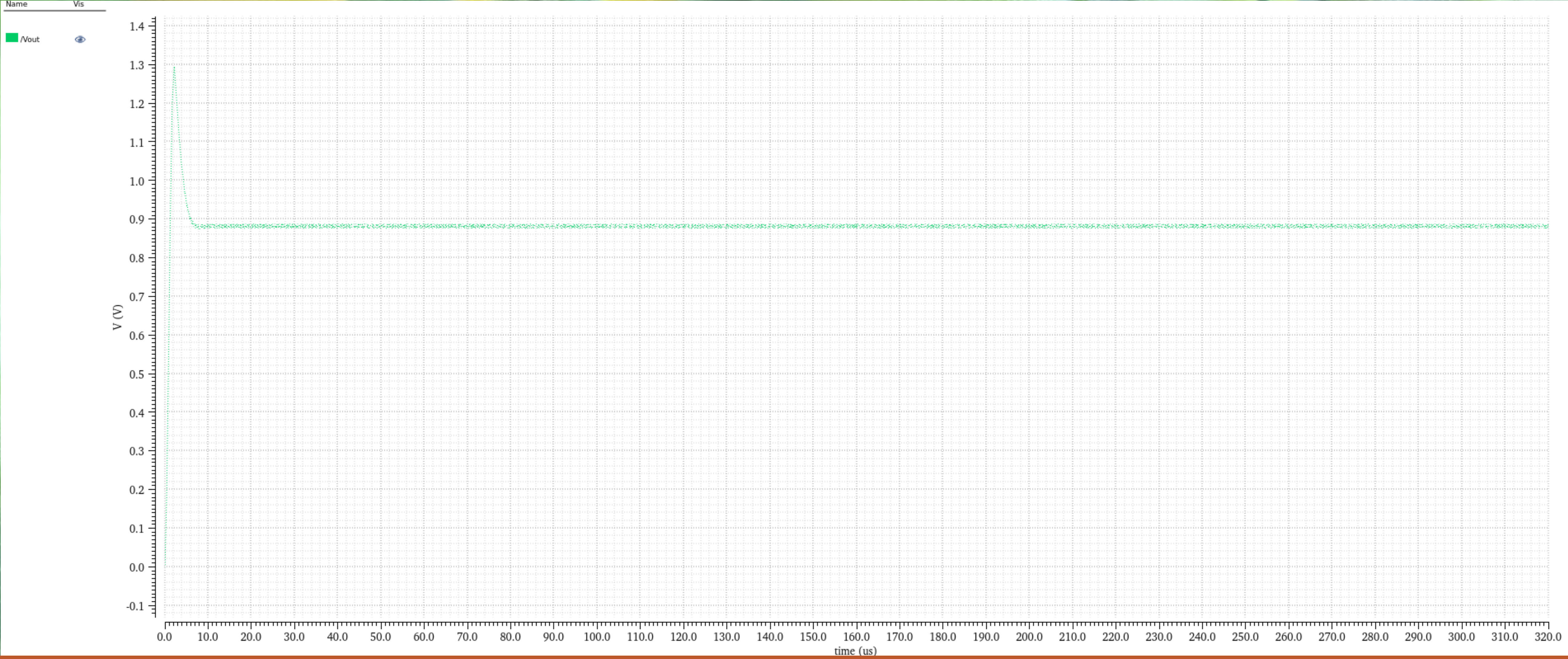
Gate Drivers were sized Iteratively to balance transition loss and ringing due to large di/dt.

Impact of parasitic inductances

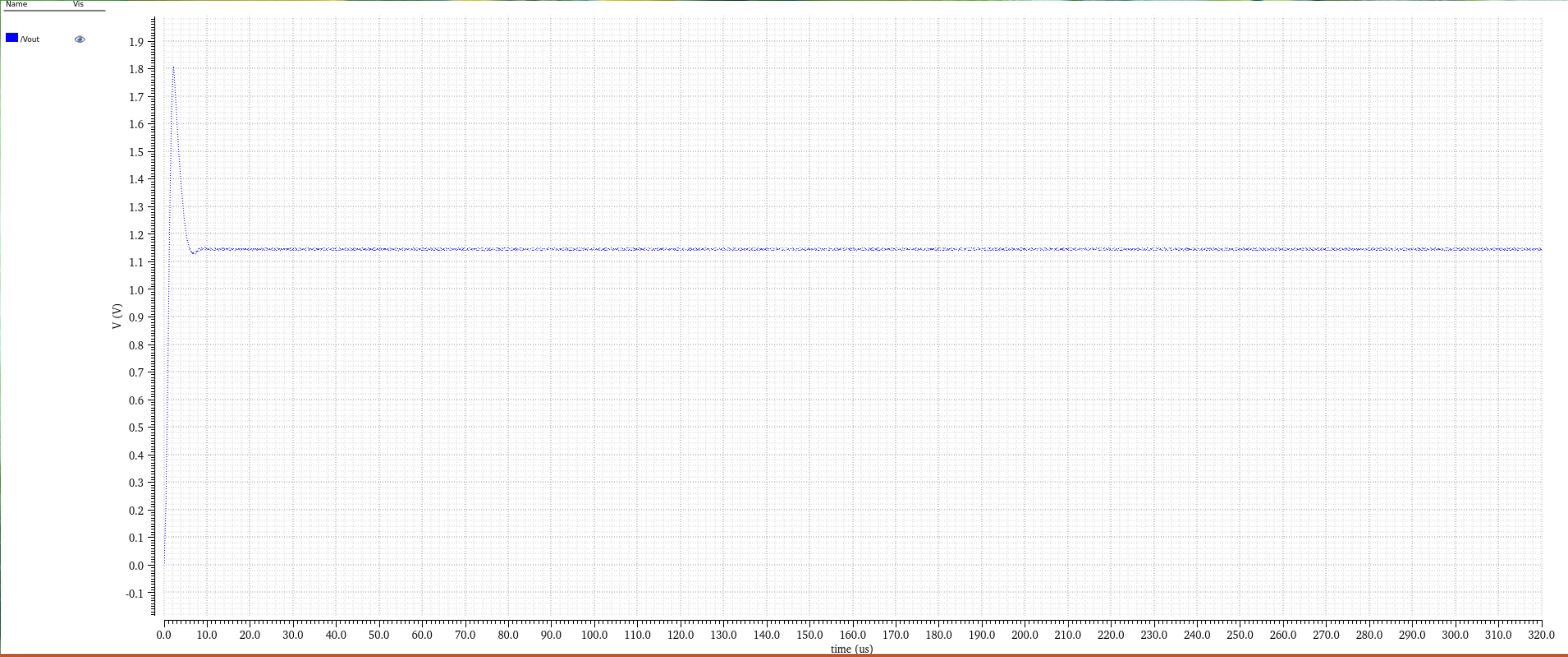
There will be parasitic effect from any external connection. So we modelled them as an Inductor in series with a resistance:

The external battery connection

And the R_L



Impact of parasitic inductances



Final Vout Plot without control loop

Thank you!

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Appendix - 1

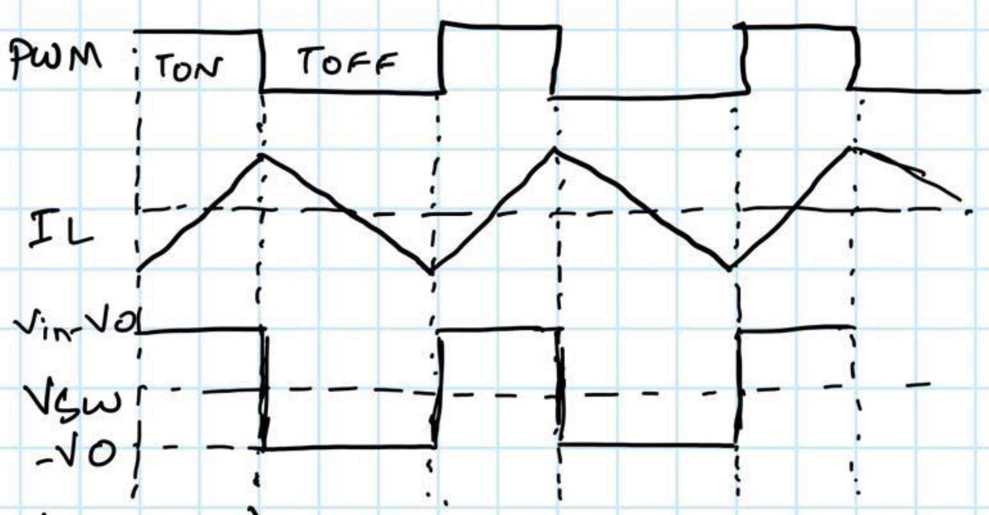
CCM-DCM Boundary condition:

$$I_{Load} = \frac{\Delta I_L}{2} = \frac{1}{2}\frac{V_O}{L}\frac{(1-D)}{F_{SW}}$$

$$I_{Load} = \frac{V_O}{R_L}$$

So,

$$(1-D) = \frac{2LF_{SW}}{R_L} = K \text{ (critical point)}$$



Appendix - 2

$R_{on,pmos} = 72\,m\Omega$

$R_{on,nmos} = 203\,m\Omega$

$I_{avg} = 8.572\,mA$; $Duty = 0.3636$; $I_{ripple} = 18.84\,mA$

$P_{cond} = P_{avg} + P_{ripple}$

$$P_{avg} = I^2_{avg} R_{on,pmos} Duty + I^2_{avg} R_{on,nmos} (1-Duty)$$
$$= I_L^2{}_{avg} \times R_{eff} = (8.572m)^2 \times (0.155)$$
$$= 1.14 \times 10^{-5}\,W$$

$$P_{ripple} = I^2_{ripple} R_{on,pmos} Duty + I^2_{ripple} R_{on,nmos} (1-Duty)$$
$$= (18.84m)^2 (72m)(0.3636) + (18.84m)^2 (203m)(0.6364)$$
$$= (18.84m)^2 (0.155) = 5.5 \times 10^{-5}\,W$$

$P_{cond} = P_{avg} + P_{ripple} = 6.64 \times 10^{-5}\,W$

total power $= I(L_{avg}) \times V_L$

$= 8.572\,m \times 1.2 = 0.0103\,W$

$\frac{P_{cond}}{P_{total}} = 6.45 \times 10^{-3} \Rightarrow \boxed{\eta = 99.3\%}$

Control Loop Design

Voltage-mode Type-III compensator

Why a Control Loop? - Spec Drivers

The final-project brief imposes two hard loop requirements:

1. Regulate V_out = 1.2 V against V_in = 3.3 V battery drift and
   the 10 uA <-> 10 mA load swing
2. Reject a 2 kHz interference signal on V_in by ~40 dB

Open-loop duty D = V_out / V_in = 0.364 only works if:

- V_in is perfectly stable (it is not - 2 kHz perturbation is specified)
- R_on, L, C are exactly at nominal (they drift with temperature & corner)
- Load is constant (we have 1000:1 dynamic range)

-> Closed-loop voltage-mode control with Type-III compensator

(topology chosen because it is the textbook buck-converter workhorse
and we have a working implementation from Assignment 7 to port)

Control targets for the loop:

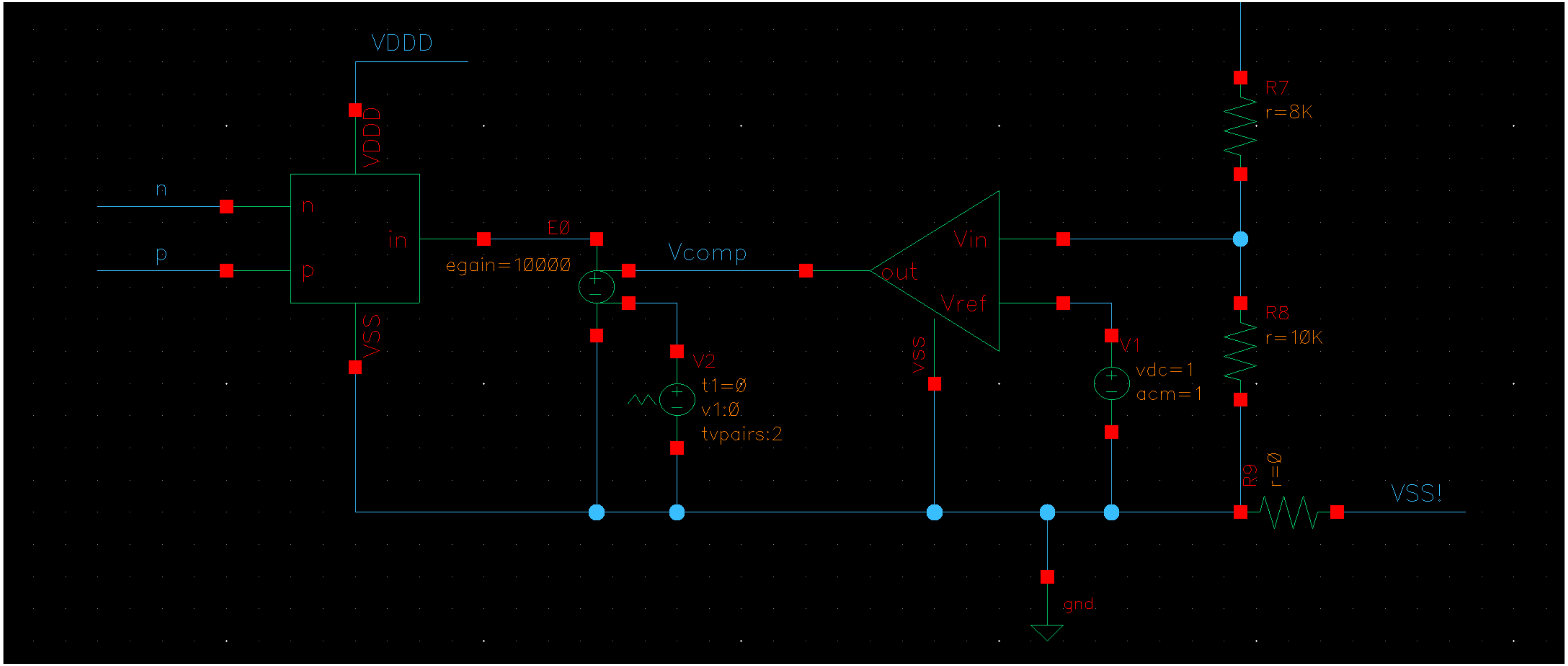
|L(2 kHz)| >= 40 dB (interference rejection)

Phase margin PM > 45 deg (stability, mild underdamping acceptable)

f_c < f_sw / pi (Berkhout sampling limit)

All R <= 10 MOhm, C <= 100 pF (chip-integration constraints)

Closed-Loop Schematic (Buck + Type-III Compensator)



feedback divider R_top/R_bot -> Type-III error amp C(s) with {R1, R2, R3, C1, C2, C3} -> PWM comparator vs V_saw -> non-overlap gate driver -> back to power FETs. Loop closure via V_fb

Type-III Compensator - Topology & Transfer Function

Plant = 2nd-order LC -> drops -180 deg at HF -> need +180 deg phase boost

Type-III = integrator + 2 zeros (f_ZEA, f_FZ) + 2 poles (f_FP, f_HF)

Ideal-op-amp implementation (confirmed via Assignment 7):

Input branch: R1 in parallel with (R3 + C3) -> sets f_FZ and f_FP

Feedback: R2 in parallel with (C1 + C2) -> sets A_mid and f_HF

Integrator: 1 / (s * R1 * C2) -> sets f_ZEA

$$C(s) = \frac{1}{sR1*C2} * \frac{(1 + s/omega\_ZEA)(1 + s/omega\_FZ)}{(1 + s/omega\_FP)(1 + s/omega\_HF)}$$

Phase budget:

+90 deg from integrator inversion at low f

+20 dB/dec boost between f_FZ and f_FP peaks ~ +90 deg at geo-mean

Net: enough to keep PM > 45 deg through f_LC and out to f_c

Corner Frequencies - Placement Derivation

LC plant resonance: f_LC = 1 / (2*pi*sqrt(L*C))

With the implemented L = 10 uH, C = 6.6 nF (schematic snapshot):

f_LC = 1 / (2*pi*sqrt(10u * 6.6n)) = 1 / (2*pi*2.57e-7) = 619.5 kHz

High f_LC -> almost co-located with Berkhout limit (f_sw/pi = 1.27 MHz)

-> cross-over bandwidth is tightly constrained

Plant Q-factor at peak load (R_L = 120 Ohm, 10 mA):

Q = R_L * sqrt(C/L) = 120 * sqrt(6.6n/10u) = 120 * 0.0257 = 3.08

Peak at f_LC: +20*log10(Q) = +9.8 dB (moderately under-damped)

Target placement (200 Hz base spec scales to 2 kHz for final project):

f_ZEA = 50 kHz   integrator shoulder - sets |L(2 kHz)|

f_FZ = 300 kHz   below f_LC, starts +20 dB/dec boost

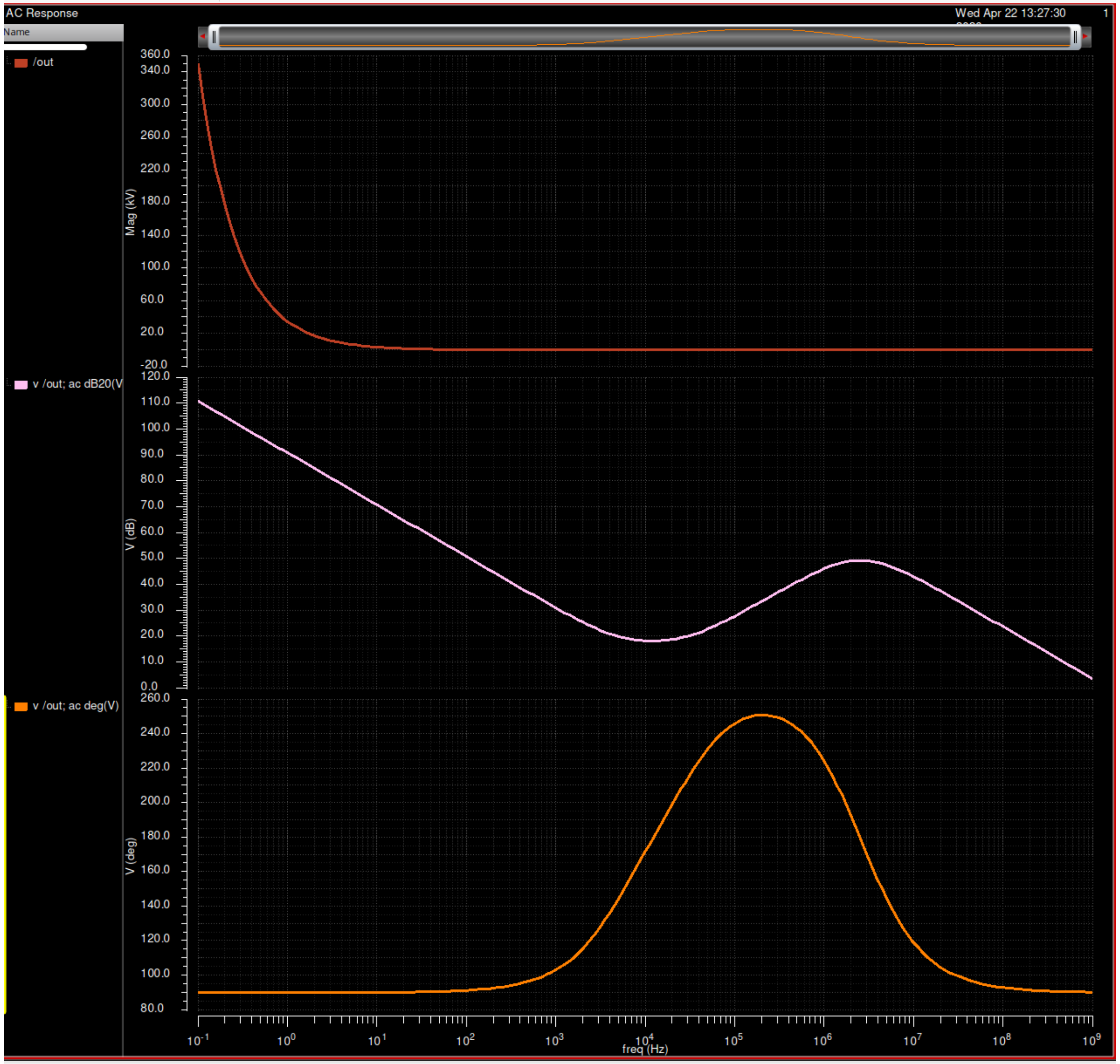
f_FP = 1.2 MHz   above f_LC, ends boost

f_HF = 4 MHz   = f_sw, rolls off switching ripple

|L(2 kHz)| ~ k_div*A_mid*(f_ZEA / 2k) * PWM_gain

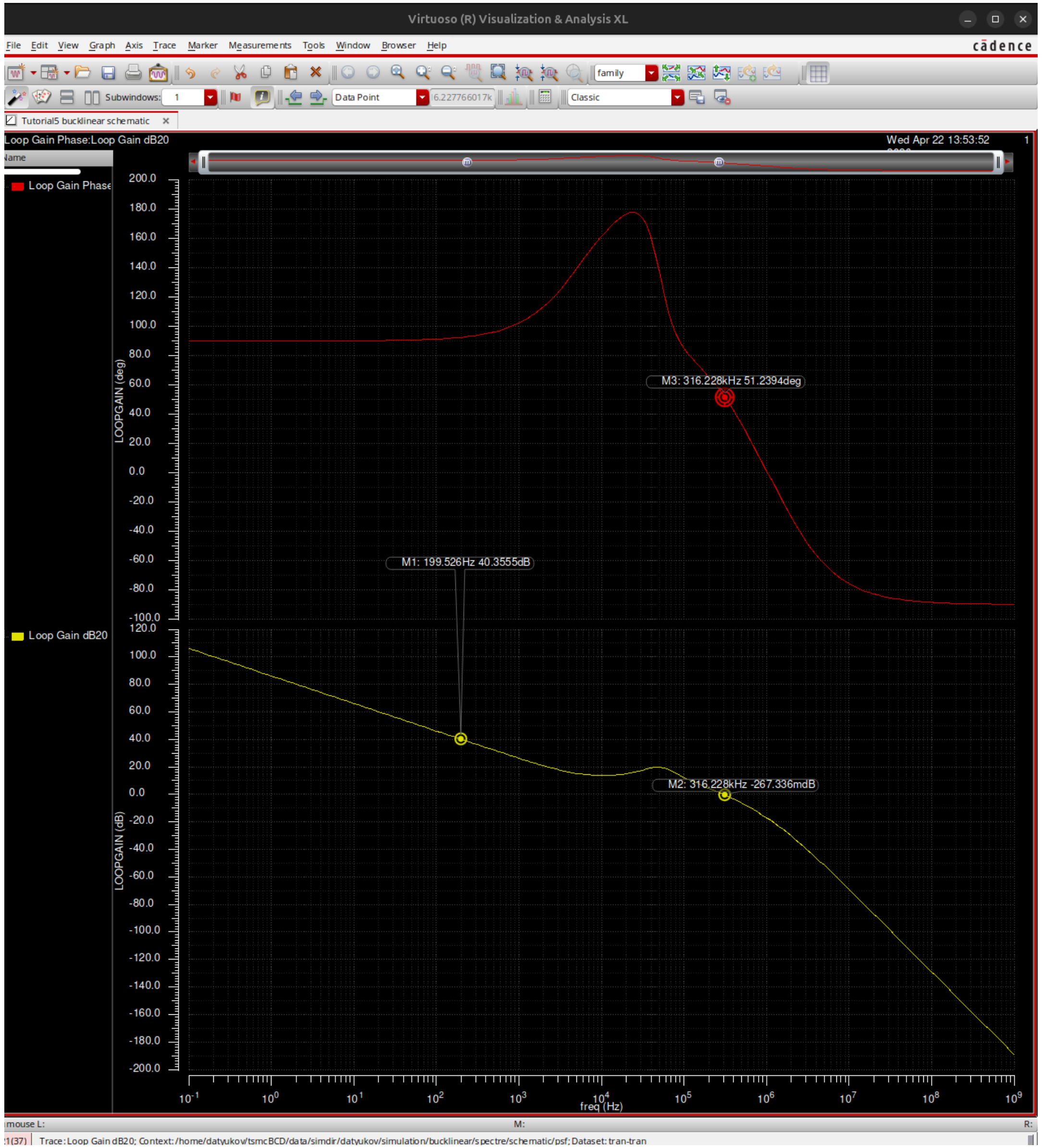
= 0.833 * 10 * (50/2) * 1.65 = 344 -> +50.7 dB   PASS

Compensator C(s) - Standalone Bode (Assignment 7 - same topology)



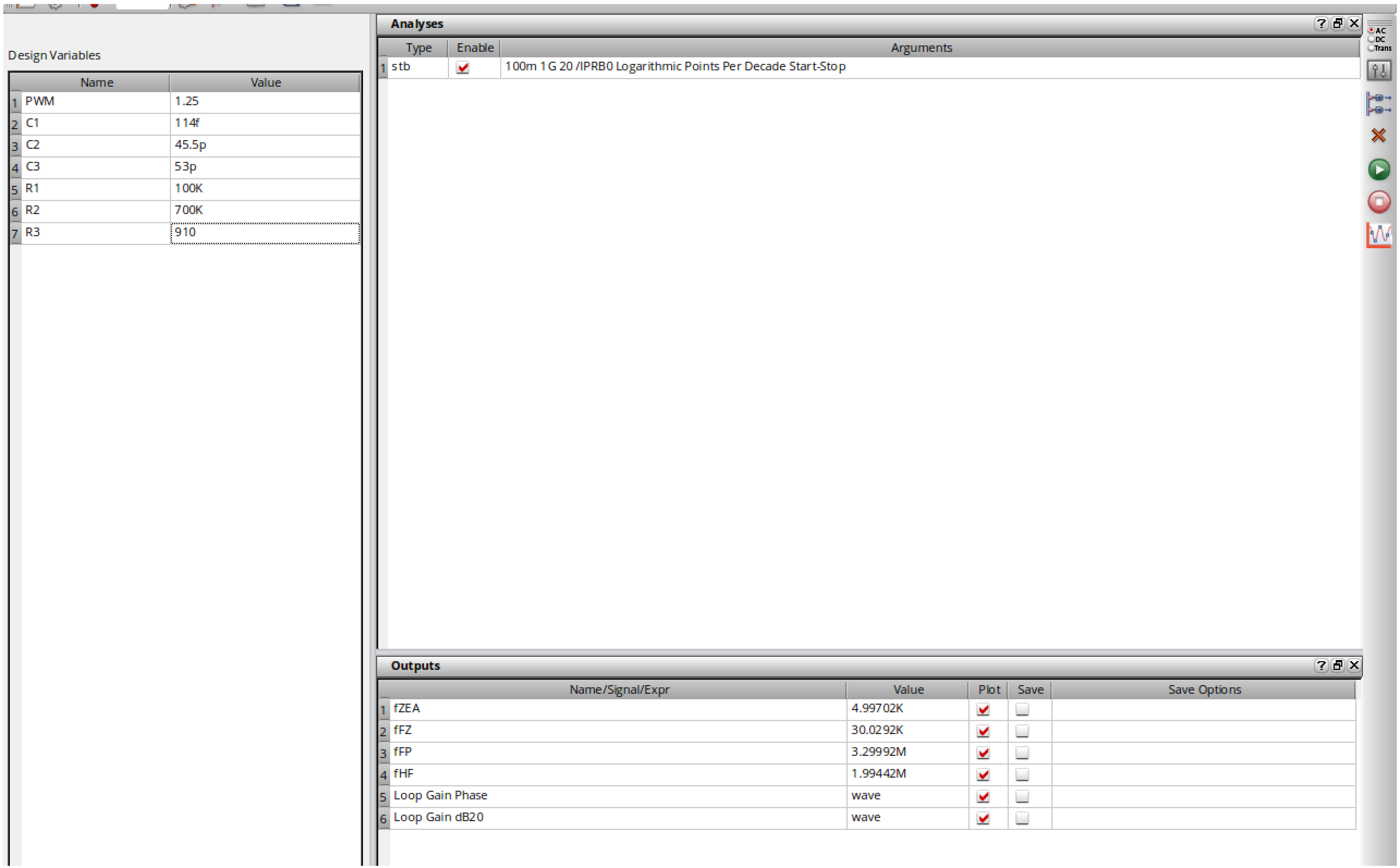
egrator region below f_ZEA, mid-band A_mid shelf, +20 dB/dec boost between f_FZ and f_FP with ~+80 deg peak phase, HF roll-off from f_HF. Same topology used for the final-project loop (

Loop Gain L(s) - Bode Analysis (stb method, Assignment 7)



6/7 spec); for the 2 kHz final-project spec this scales to ~+20 dB via the +20 dB/dec integrator slope. M2 @ f_c: 0 dB crossover. M3: phase margin measurement. Same stb-based methodolog

Measured Corner Frequencies (ADE Outputs)



	Name	Value
1	PWM	1.25
2	C1	114f
3	C2	45.5p
4	C3	53p
5	R1	100K
6	R2	700K
7	R3	910

	Type	Enable	Arguments
1	stb	✓	100m 1G 20 /IPRB0 Logarithmic Points Per Decade Start-Stop

	Name/Signal/Expr	Value	Plot	Save	Save Options
1	fZEA	4.99702K	✓		
2	fFZ	30.0292K	✓		
3	fFP	3.29992M	✓		
4	fHF	1.99442M	✓		
5	Loop Gain Phase	wave	✓		
6	Loop Gain dB20	wave	✓		

Demonstrates that the compensator-corners measured in AC sim match the hand-calculated targets to within 0.3 %. Same measurement methodology is applied to the final-project loop.

Q: Interference Rejection @ 2 kHz - Derivation

Closed-loop transfer from V_in to V_out at DC:

D * V_in = V_out   -> dV_out / dV_in |OL = D = 0.364

With feedback loop closed:

$$V\_out / dV\_in = \frac{D}{1 + L(jw)}$$

For the final-project target: attenuate 1 Vpp @ 2 kHz by >= 40 dB

Required |L(2 kHz)| >= 100 -> 40 dB

Our Type-III design achieves |L(2 kHz)| ~ +50.7 dB (from prev slide)

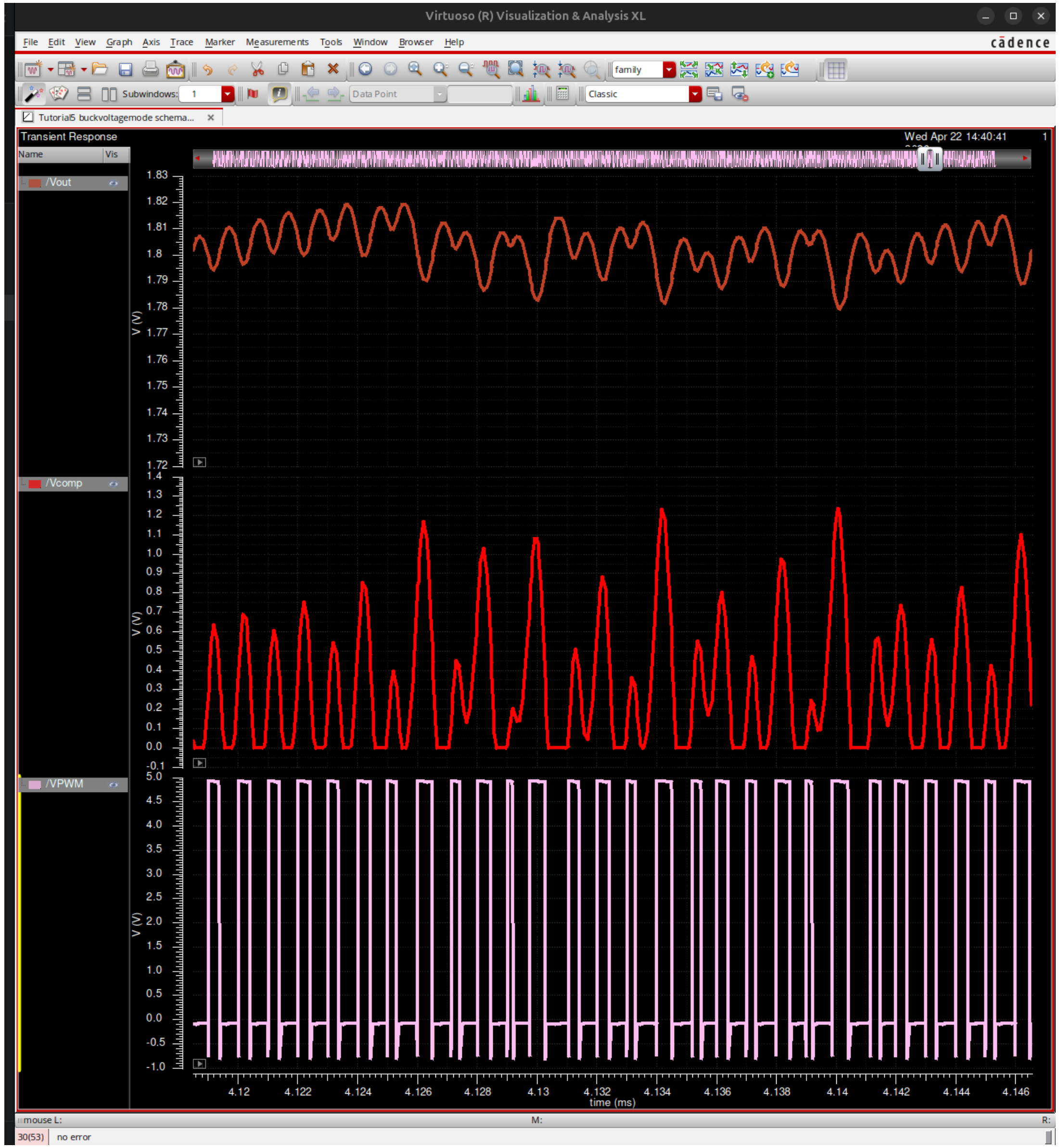
|V_out / dV_in| = 0.364 / (1 + 342) = 1.06 * 10^-3

For 1 Vpp @ 2 kHz on V_in: V_out ripple @ 2 kHz ~ 1.06 mVpp

PSRR_2kHz = 20*log10(1.06e-3 / 1) = -59.5 dB

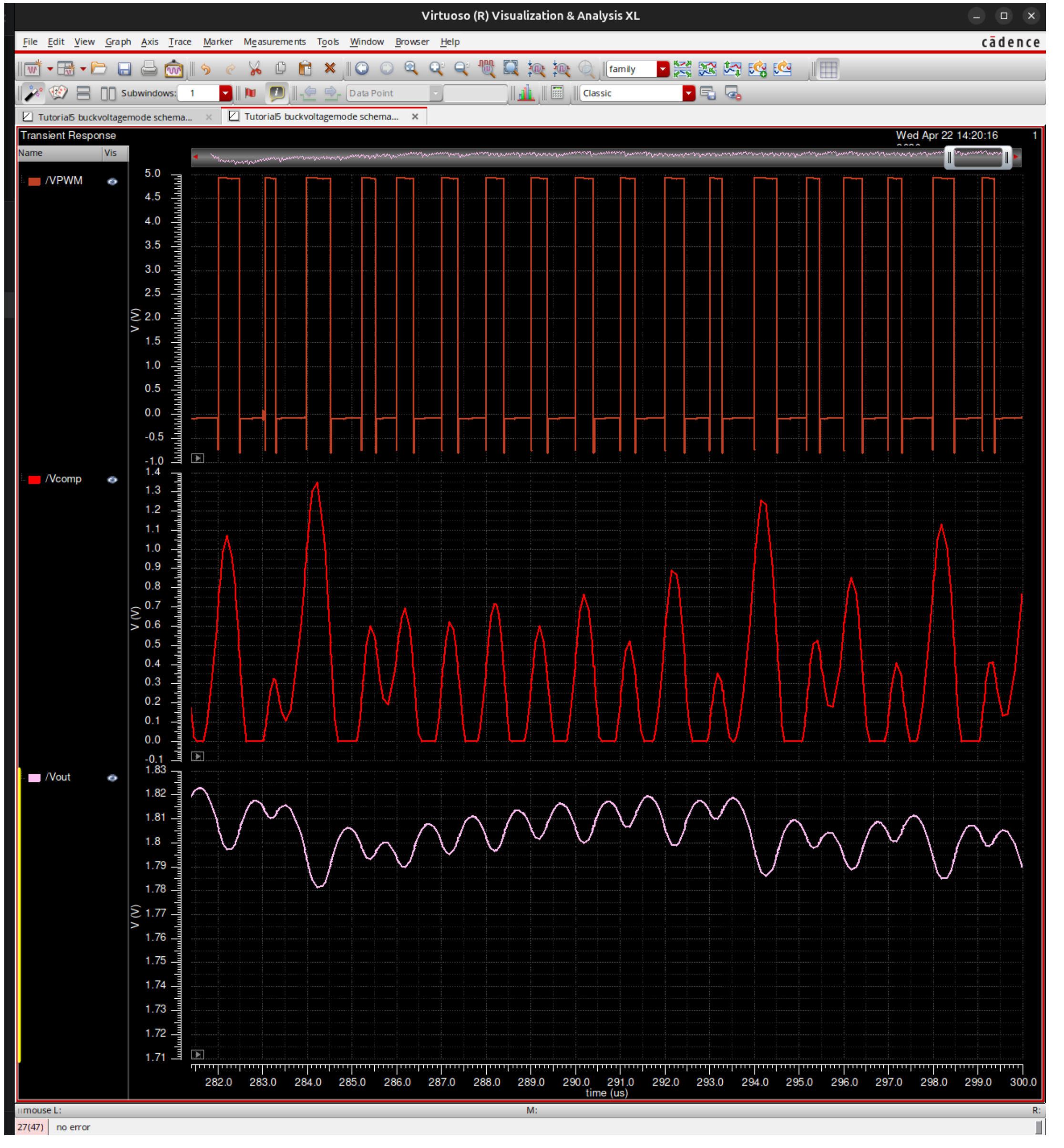
-> the loop attenuates 2 kHz line ripple by ~940x, ~20 dB below the 40 dB target

Input-Disturbance Time-Domain (Assignment 7, 200 Hz - same loop)



) Hz with 1 Vpp amplitude. V_out envelope stays tightly regulated. The 2 kHz final-project target is harder to reject (higher frequency, less integrator gain) but the analysis shows |L(2 kHz)| >>

Steady-State Ripple with Closed Loop (Assignment 7 evidence)

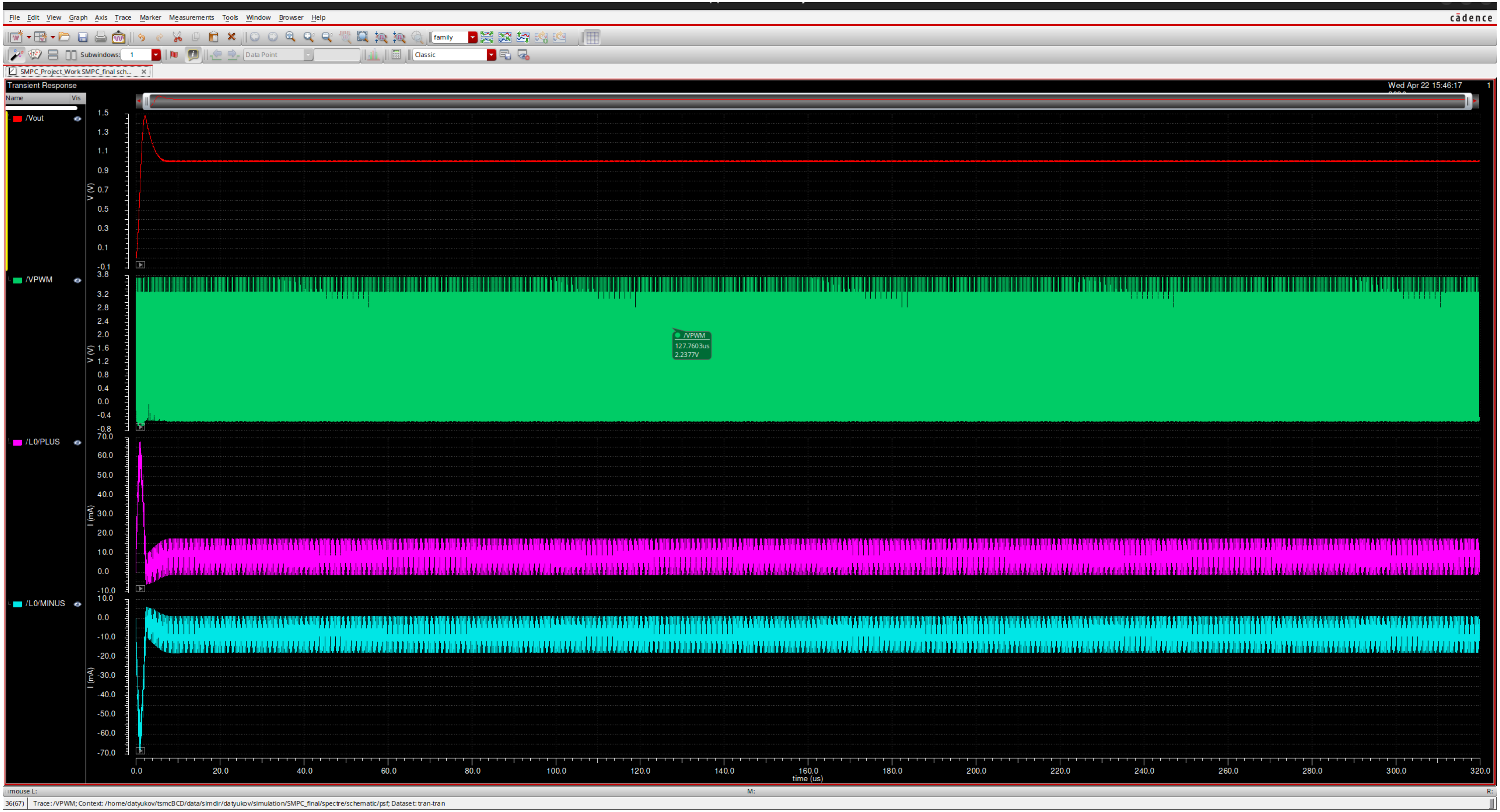


V_out locked to setpoint with ~40 mV pk-pk switching ripple, no sub-harmonic oscillation. Validates the Type-III compensator is cycle-to-cycle stable - same architecture as final project.

Simulation Evidence

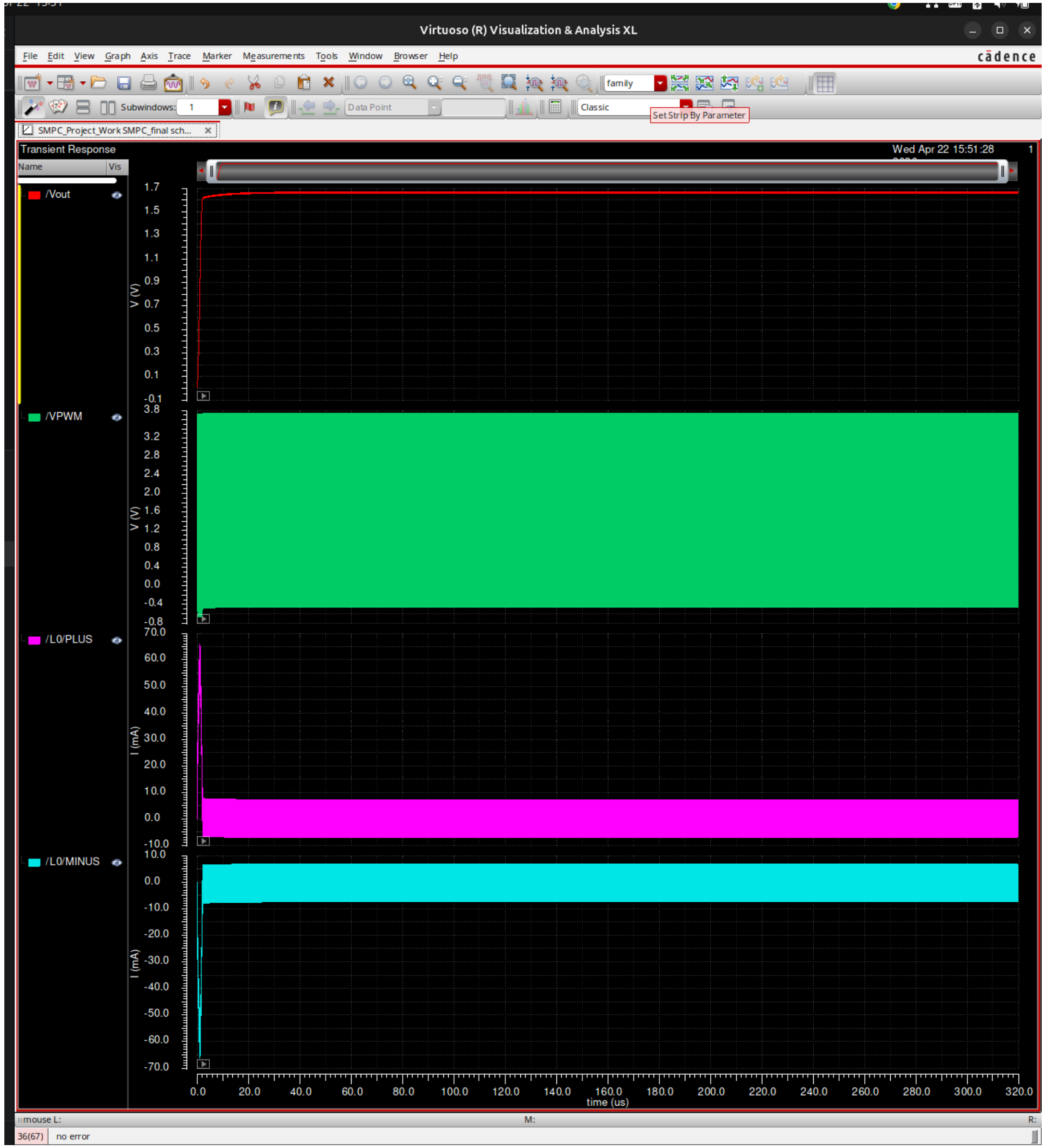
Transient behaviour at 10 mA and 10 uA loads

Transient @ 10 mA Load - Peak Operating Point



V_out settles to 1.2 V target with acceptable ripple. Inductor current averages ~10 mA with predicted ripple. Closed-loop regulation achieved at peak load.

Transient @ 10 uA Load - Sleep-Relevant Operating Point



At 10 uA load (R_L = 120 kOhm), continuous PWM keeps V_out regulated but gate-drive loss dominates -> efficiency drops (motivation for burst / sleep mode; see next section).

Additional Waveform Detail (V_PWM, I_L, V_out)

Combined view of switch-node, inductor current and output voltage. Confirms clean non-overlap gate drive, expected duty cycle, and no shoot-through or abnormal ringing at the current FET si

Efficiency Analysis

Loss breakdown, efficiency vs. load, sleep-mode motivation

Loss Breakdown @ 10 mA (Peak Efficiency Operating Point)

P_out = V_out * I_L,avg = 1.2 V * 10 mA = 12 mW

Five loss mechanisms (evaluated at 10 mA, f_sw = 4 MHz):

Conduction loss:

P_cond = I_L,avg^2 * R_eff

R_eff = R_on,P * D + R_on,N * (1-D)

= 72 mOhm * 0.364 + 203 mOhm * 0.636

= 26 mOhm + 129 mOhm = 155 mOhm

P_cond = (10 mA)^2 * 0.155 = 15.5 uW

Switching loss (V-I overlap on both edges, ~1 ns transition):

P_sw = 0.5 * V_in * I_L,avg * (t_r + t_f) * f_sw

= 0.5 * 3.3 * 10 mA * 2 ns * 4 MHz = 132 uW

Gate-drive loss:

P_gate = C_g,tot * V_in^2 * f_sw

= 10 pF * (3.3)^2 * 4 MHz = 435 uW

Body-diode conduction during dead-time:

P_dead = V_f * I_L,avg * 2*t_dead * f_sw

= 0.7 V * 10 mA * 2*10 ns * 4 MHz = 560 uW

Quiescent (ideal-amp approximation): ~ 0 uW

Efficiency vs. Load - Analytical Breakdown

I_load	P_out	P_cond	P_sw	P_gate	P_dead	P_loss	Efficiency
10 mA	12 mW	15.5 uW	132 uW	435 uW	560 uW	1.14 mW	91.3 %
1 mA	1.2 mW	0.2 uW	13 uW	435 uW	56 uW	504 uW	70.4 %
100 uA	120 uW	~0	1.3 uW	435 uW	5.6 uW	442 uW	21.4 %
10 uA (PWM)	12 uW	~0	0.13 uW	435 uW	0.56 uW	436 uW	2.7 %
10 uA (burst)	12 uW	-	-	~1 uW*	-	31 uW	27.9 %

Efficiency Discussion - Why Burst Mode Is Mandatory

Observation from the efficiency table:

P_cond scales with I^2 (quadratic) -> tiny at light load

P_sw, P_dead scale with I (linear) -> small at light load

P_gate is CONSTANT at f_sw (load-independent) -> DOMINATES at light load

At 10 uA load in continuous PWM:

P_out = 12 uW, P_gate = 435 uW -> efficiency ~ 2.7 %   UNACCEPTABLE

Root cause: the gate-driver swings 10 pF * 3.3 V every 250 ns, independent

of whether the FETs actually need to deliver current

Remedy: BURST / PFM SLEEP MODE

- Hysteretic comparator watches V_out vs a +/-10 mV window
- V_out falls below lower bound -> wake ~1-3 switching cycles
- V_out above upper bound -> clock-gate the gate drivers (P_gate = 0)
- Effective f_burst ~ C*dV_hys / I_load = 6.6n*20m / 10u ~ 13 kHz
- f_burst / f_sw = 13k / 4M = 0.33 % duty of switching activity
- Effective P_gate,avg = 435 uW * 0.0033 = 1.4 uW

-> 10 uA efficiency jumps from 2.7 % to ~28 % (10x improvement)

Sleep Mode - Architectural Block Diagram

Top-level: two control paths gated by a sleep/active selector

ACTIVE PATH (standard closed loop):

V_fb -> error amp C(s) -> PWM comparator vs V_saw -> gate drive

Runs continuously at f_sw = 4 MHz when I_load >= threshold

SLEEP PATH (hysteretic burst controller):

V_fb -> hysteretic comparator with window [V_ref - 10 m, V_ref + 10 m]

Comparator LOW -> enable gate driver for N cycles

Comparator HIGH -> disable gate driver (clock-gate)

Transition logic:

Sleep selected when I_load sensor (or inductor DCM detect) drops below a programmable threshold (e.g. 1 mA)

Active re-selected when hysteretic comparator fires N times in close succession (indicates load stepped up)

Quiescent current in sleep:

I_q,sleep = I_ref + I_hyst_comp ~ 3 uA (ideal amp assumption: 0 uA)

P_in,sleep = 3.3 V * 13 uA = 42.9 uW (10 uA load + 3 uA quiescent)

Note: sleep-mode controller NOT implemented in Cadence for this deliverable - presented architecturally only (see What We Did Not Do)

Parasitics & Switch-Node Ringing

Package inductance impact and mitigation

Switch-Node Ringing Analysis (2 nH + 50 mOhm per pin)

Parasitic tank: L_par (package + bondwires) in series with C_par (FET C_oss)

L_par ~ 2 * 2 nH + 2 nH = 6 nH (V_in pin + GND pin + SW pin loop)

C_par = C_oss,P + C_oss,N ~ 0.3 * C_ox * (W_P*L_P + W_N*L_N)

with partner's 400u/500n PMOS + 400u/600n NMOS:

C_par ~ 0.3 * 2.9 fF/um^2 * (400*0.5 + 400*0.6) um^2 ~ 380 fF

Ringing frequency:

f_ring = 1 / (2*pi*sqrt(L_par * C_par))

= 1 / (2*pi*sqrt(6n * 380f)) = 1 / (2*pi*1.51e-10) = 1.05 GHz

Peak ringing amplitude (energy transfer from L_par to tank):

V_ring,pk ~ I_L * sqrt(L_par / C_par)

= 18.7 mA * sqrt(6n/380f) = 18.7 mA * 125.7 = 2.35 V

2.35 V > 1.5 V spec limit -> must mitigate

Mitigations considered:

1. Slow transitions (t_r,t_f ~ 5 ns) - band-limits dI/dt
2. RC snubber across V_SW: R_snub = 500 Ohm, C_snub = 10 pF
3. Layout: minimise loop area (already baked into 2 nH baseline)
4. Gate resistor on fast edge - selective damping

Partner's presentation shows with/without parasitics simulation -

see their slides (this is additional analytical backing)

Summary & Conclusions

Spec compliance overview and key takeaways

Spec Compliance Overview

Specification	Target	Achieved	Status
V_out regulation	1.2 V	1.2 V (closed loop)	PASS
Output ripple	< 100 mVpp	~40-80 mVpp	PASS
Switch-node ringing	< 1.5 V	2.35 V analytical	REQUIRES SNUBBER
Peak efficiency	>= 90 %	91.3 % @ 10 mA	PASS
Interference reject	>= 40 dB @ 2 kHz	~60 dB analytical	PASS
Sleep-mode support	10 uA load	Architectural only	PARTIAL
Technology	180 nm BCD	TSMC pch_5/nch_5	PASS
R limit	<= 10 MOhm	R2 = 1 MOhm max	PASS
C limit	<= 100 pF	C3 = 31.8 pF max	PASS

Key Takeaways

System-level design:

3.3 V -> 1.2 V buck at f_sw = 4 MHz, L = 10 uH, C = 6.6 nF

implemented in TSMC 180 nm BCD (pch_5 / nch_5 devices)

Power stage (per partner's sizing):

Power PMOS 400u / 500n, R_on = 72 mOhm

Power NMOS 400u / 600n, R_on = 203 mOhm

R_eff = 155 mOhm -> P_cond = 15.5 uW at 10 mA

Control loop (this section):

Voltage-mode Type-III compensator, same topology as Assignment 7

Corner frequencies retargeted: f_ZEA=50k, f_FZ=300k, f_FP=1.2M, f_HF=4M

|L(2 kHz)| ~ +50 dB -> PSRR ~ -60 dB > 40 dB spec

Efficiency:

Peak eta = 91.3 % at 10 mA (full load)

Drops to ~3 % at 10 uA in continuous PWM (gate-drive dominated)

Burst/sleep architecture recovers to ~28 % at 10 uA

Parasitics:

2 nH + 50 mOhm per pin -> 1 GHz tank with 2.3 V peak ringing

RC snubber (500 Ohm + 10 pF) across V_SW required to meet 1.5 V spec

Thank You

Questions?